Phase 1, Module 6: Control Flow & System Automation

Khyber AI Systems Engineer (KASE)

Welcome to Module 6. Until now, our code runs top-to-bottom and stops. Today, we give our systems the ability to make decisions and automate repetitive tasks. This is where your code becomes a 'System’

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Conditionals (The System Router)

Python 3.4.1: dogs litter number.py - //vmware-host/Shared Fo

File Edit Format Run Options Windows Help

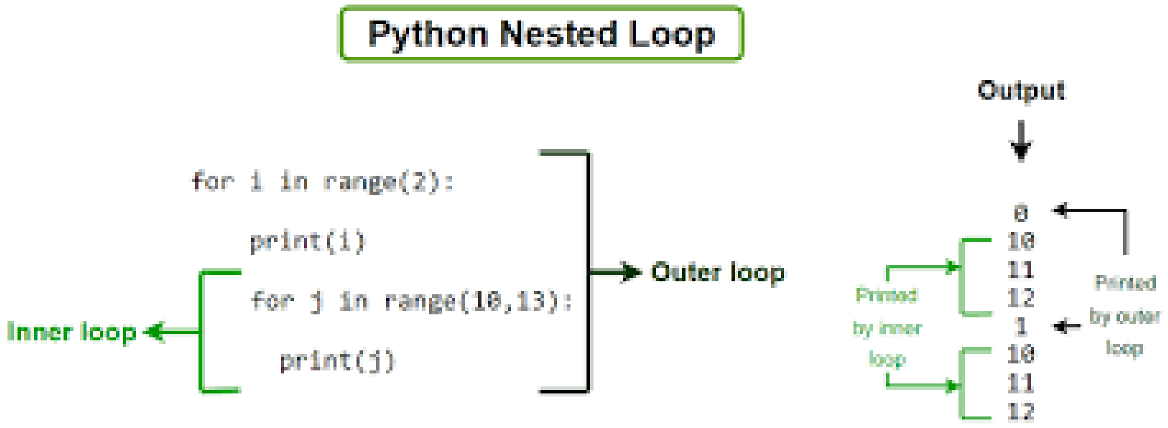
```python
litter = int(input("How many puppies were born?"))

if litter <= 5:
    print("good size")
elif litter == 6:
    print("just right")
elif litter == 7:
    print("large litter")
else:
    print("goodness me")
```

$if, elif, else$

- The brain's routing mechanism.
- if: The primary check.
- elif: The secondary check (only runs if the first fails).
- else: The default fallback (prevents system crashes).

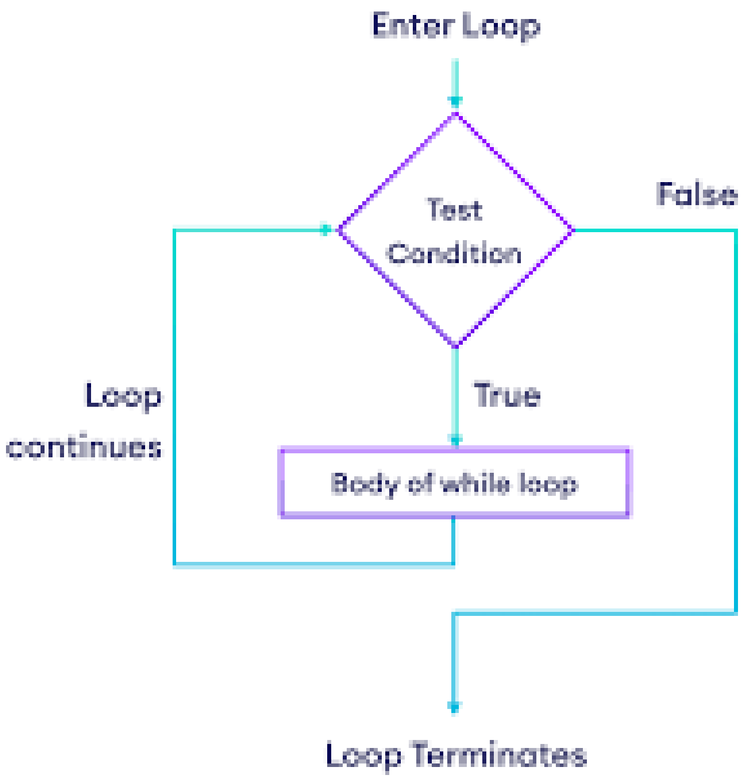
The for Loop (Batch Processing)



Predictable Automation

- Use a for loop when you know exactly how much data you have.
- Perfect for iterating through Lists (datasets) and Dictionaries (JSON).

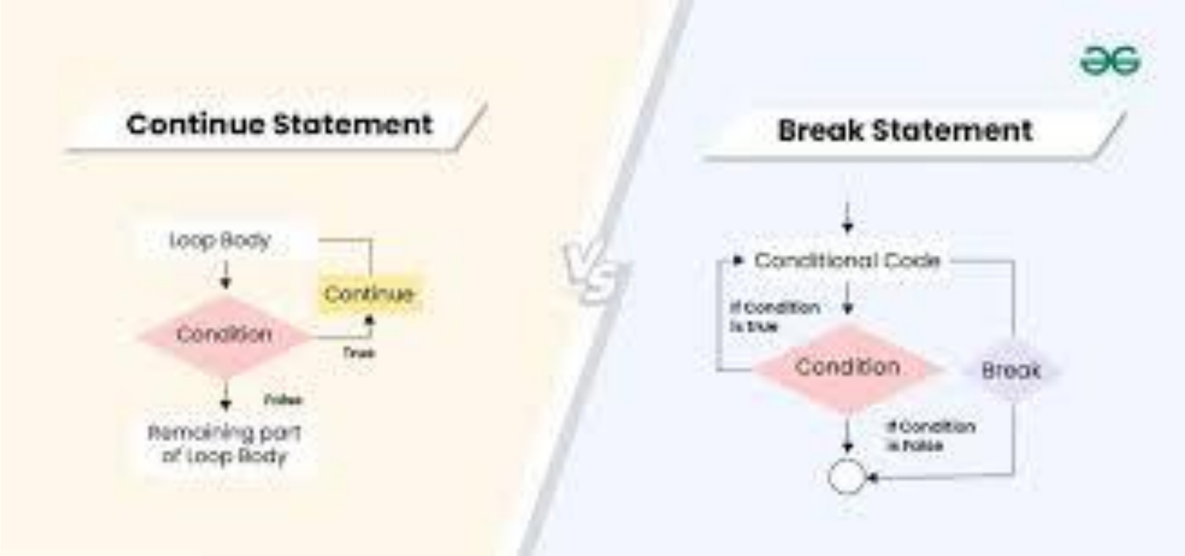
The while Loop (The Server Engine)



Unpredictable Automation & Retries

- Use a while loop when you don't know when the task will end.
- Danger: Infinite loops will crash your RAM.
- AI Use Case: "Retry connecting to the database while the connection is False."

Compute Efficiency (*break* & *continue*)



Saving GPU Cycles

- ***continue***: Skip this specific item, but keep the loop running. (e.g., skip a corrupted image).
- ***break:*** Kill the loop entirely. (e.g., Target found, stop searching).